# AquaSentinel R3: 8-minute semi-final pitch

India Semi-Final, 24-25 Oct 2026 · deck: AquaSentinel_SemiFinal_FINAL.pptx (the same words are in its speaker notes) · about 140 words a minute; about 880 spoken words plus the 60 s video = about 7:30, leaving 30 s for pauses and clicks

| Slide | Time | Speaker | Topic |
|---|---|---|---|
| 1 | before the timer | Soham | Cover |
| 2 | 0:00-0:40 | S1 Soham | The problem |
| 3 | 0:40-1:45 | S1, then the video | Marketing video |
| 4 | 1:45-2:25 | S2 Tanmay or Bhakti | What it is |
| 5 | 2:25-3:10 | S2 | How it works |
| 6 | 3:10-3:50 | S3 (whoever knows the CAD best) | Design checks and simulation |
| 7 | 3:50-4:20 | S3 | Materials and manufacturing |
| 8 | 4:20-5:30 | S1 Soham | Customers, cost and business model |
| 9 | 5:30-6:15 | S2 | Where we are, honestly |
| 10 | 6:15-6:45 | S1 Soham | Prior art |
| 11 | 6:45-7:15 | S1 Soham | How AquaSentinel differs |
| 12 | 7:15-7:50 | S1 Soham | Close |

**Rules we keep:** say "designed", not "built"; "a sample for the lab", never "evidence"; quote safety factor 2.66 (static case), not 4.4; say "synthetic data" with the 4-of-4 result; never quote IDC/FDC grades. Q&A menu: type 13 and press Enter. Prior art is slides 10-11; backups A1-A9 are slides 14-22.

## Slide 1 · Cover before the timer · Soham

*Show the cover while we are introduced. When the 8-minute timer starts, click once to slide 2. Animations play by themselves. In Q&A, type 13 and press Enter for the questions menu (slides 10-11 are prior art; 14-22 are backups A1-A9).*

## Slide 2 · The problem 0:00-0:40 · S1 Soham

> It's two in the morning on the Godavari. A tanker opens its valve, and by sunrise the river looks normal again. Nobody saw it, and the next official sample may be weeks away. India's latest CPCB assessment lists 296 polluted river stretches and locations, and Maharashtra, our state, has the most: 54. Most monitoring points are sampled by hand, monthly or quarterly. Real-time stations exist, but they cost tens of lakhs each, so there are very few. A spill that lasts a few hours falls through the gap. We are Team Sanjivani Hero's Journey, and AquaSentinel is our answer.

## Slide 3 · Marketing video 0:40-1:45 · S1, then the video

*Say: 'Here is AquaSentinel in sixty seconds.' Click the video to play it. If it does not start, play AquaSentinel_Marketing_60s.mp4 from 02_PRESENTATION_SEMIFINAL and keep Zoom's Share sound on.*

### Slide 4 · What it is 1:45-2:25 · S2 Tanmay or Bhakti

> AquaSentinel is a river robot designed to sense, think and act on its own. It senses pH, conductivity, dissolved oxygen, turbidity, temperature and river level. An ESP32 on board decides every fifteen minutes, with no cloud needed. When the water changes, the beacon turns red, an alert goes to the water-works, and a pump fills a sample bottle for the lab. It is an early-warning system, not a chemical identifier. The table shows each objective, and honestly, how far it is proven.

### Slide 5 · How it works 2:25-3:10 · S2

> How does it stay in place without a motor? A two-inch stainless pole is clamped to a bridge pier, so nobody works in the riverbed. Eight rollers in two tiers let the river itself carry the float up and down through two metres, and its side area sits behind the pole, so the current turns it like a weathervane. Every fifteen minutes it checks hard limits, like pH outside six to nine or oxygen below four, and a baseline it learns for every hour of the day. Two unusual readings out of three make an event.

### Slide 6 · Design checks and simulation 3:10-3:50 · S3 (whoever knows the CAD best)

> We checked every requirement on the CAD model itself. The float weighs 48.5 kilograms by our estimate, floats with 143 millimetres of freeboard and more than twice the buoyancy it needs, and rights itself up to sixty degrees. For a flood we designed for 1.5 metres per second plus a 500-newton debris load. Simulation with fixed ends gives a safety factor of 4.4; we quote our most conservative case, 2.66. That is a static case: impact and anchors are still to verify.

### Slide 7 · Materials and manufacturing 3:50-4:20 · S3

> R3 is a 66-part SOLIDWORKS assembly with a real material on every part: two polyethylene hulls from one rotomould, aluminium deck and carriage, a stainless pole, and off-the-shelf electronics, so a local fabricator can build it. CAD also caught our two biggest mistakes: the pole once ran through the electronics pod, and the real mass came out at 48.5 kilograms, not 25.

### Slide 8 · Customers, cost and business model 4:20-5:30 · S1 Soham

> Who pays? Pollution control boards, water-works protecting their intakes, irrigation departments and industrial estates. We priced all 66 parts from Indian suppliers: one float costs about 1.6 lakh rupees before GST, plus a 29-thousand-rupee bridge kit; at a hundred units, about 1.5 lakh together. But a board doesn't buy it. It pays twelve

> thousand rupees a month on a five-year contract, like station maintenance tenders. That pays back the hardware in about twenty months, with a margin of about 46 percent. One real-time station's instruments alone cost thirty-five to forty lakh. We start with three floats at Kopargaon, then Maharashtra's 54 polluted sites.

### Slide 9 · Where we are, honestly 5:30-6:15 · S2

> Where are we, honestly? AquaSentinel is designed, calculated and simulated; it has not been in the water yet. On thirty days of synthetic data, our logic flagged all four injected spills within thirty minutes. That proves the logic, not field accuracy. Power, including the beacon, is about twelve days without sun, an estimate we will measure. Before the finale: the 1:5 model in water and the logic on an ESP32 with a real probe. Then a full-size tank test, and after permits, a thirty-day pilot at Kopargaon checked against lab samples.

### Slide 10 · Prior art 6:15-6:45 · S1 Soham

> We are not the first to watch rivers. CPCB's real-time stations read seventeen parameters every fifteen minutes. YSI's sonde is the reference instrument, NexSens floats sondes on solar buoys, and Teledyne ISCO's samplers can be triggered by a sonde. Each does its job well. But each does one part of the job.

### Slide 11 · How AquaSentinel differs 6:45-7:15 · S1 Soham

> Our difference is how we combine them: a float on a pier-mounted guide pole that follows the river without a mooring, a decision made on board, a light people on the bank can see, and one sample bottle per event, in one unit under two lakh. Each capability exists separately; the combination is what we must now prove in water.

### Slide 12 · Close 7:15-7:50 · S1 Soham

> A spill lasts hours. A sample comes once a month. AquaSentinel is designed to close that gap: ninety-six readings a day, a decision on the spot, a red light on the bank, and a sample that tells the lab where to look. We are Soham, Shriraj, Bhakti and Tanmay, guided by Dr. Kiran Wakchaure. Thank you. We'd love your questions.